

# LOKESH GUNDOJU

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## SUMMARY

Electronics & Communication Engineering student specializing in Computational Lithography, Yield Analytics, and Machine Learning for Advanced Semiconductor Manufacturing. Experienced in the numerical modeling of sub-wavelength lithography physics, time-series Statistical Process Control (SPC), and edge-ML system design, with a proven focus on data-driven process optimization and open-source EDA workflows.

## TECHNICAL SKILLS

- Process & Yield: Statistical Process Control (SPC), Yield Analytics, Computational Lithography, EUV/DUV Modeling, Root-Cause Analysis, EWMA/CUSUM, DOE.
- Software & AI: Python, C/C++, XGBoost, Ensemble ML, FastAPI, SHAP (XAI), Time-Series Analytics, Docker, Git.
- Hardware & Systems: Verilog/RTL, OpenLane, SkyWater 130nm PDK, TinyML, Altium PCB Design, Electrical Characterization.

## KEY PROJECTS

- **AttoFab** – Computational Lithography & Fabrication Simulator

Engineered an open-source, Python-based computational simulation engine targeting sub-wavelength fabrication and pattern transfer profile validation. Implemented numerical modeling of the Rayleigh Criterion for resolution limits ( $R=k_1 \cdot \lambda/NA$ ) to optimize the  $k_1$  process factor for sub-7nm process development. Simulated etching dynamics and photoresist exposure under 13.5nm Extreme Ultraviolet (EUV) and 193nm immersion conditions to match industry-standard benchmarks.

Skills Applied: Semiconductor Fabrication, Numerical Modeling, Python, Process Simulation, Research Methodology.

- **AttoSense** – Multimodal intent classifier

Accomplished production-ready NLU for intent analysis as measured by successful zero-shot discovery and multimodal auditing, by architecting a FastAPI microservice utilizing Groq Whisper (Audio) and Llama-3.2-Vision (Image).

Skills Applied: Multimodal AI, NLU, FastAPI, Microservices, Python, Computer Vision.

- **Statistical Process Control (SPC)** – Wafer Yield Analytics

Engineered an automated yield excursion monitoring framework using time-series analytics on production-grade wafer datasets, modeling a potential 20–40% reduction in defective starts. Designed baseline-calibrated EWMA and CUSUM control charts achieving a 90% anomaly detection rate and 5% false-alarm rate, shortening simulated root-cause analysis lag to 2 hours to mitigate process drift before yield degradation occurs.

Skills Applied: Statistical Process Control, Real-time Anomaly Detection, Process Optimization, Manufacturing Metrics Analysis.

- **SECOM Sensor Processing** – Yield Optimization & Machine Learning

Achieved 90% accuracy in semiconductor wafer defect classification to prevent process drift on 1,567 manufacturing runs. Optimized 8 ensemble machine learning models using stratified cross-validation to manage high-dimensional and imbalanced sensor arrays. Obtained a 95% ROC-AUC and 95% precision, providing a structured data-driven approach to yield management.

Skills Applied: Machine Learning, Ensemble Methods, Imbalanced Data Handling, Model Evaluation & Optimization, Manufacturing Process Understanding.

- **Embedded IoT & TinyML Inference Engine (CAPHA)**

Designed an autonomous, real-time inference framework for low-power microcontrollers featuring proactive anomaly detection. Integrated TinyML models onto IoT edge devices, enabling autonomous hardware decisions and low-latency local processing. Maintained a minimal RAM/Flash footprint to run high-fidelity signal analysis without external cloud reliance.

Skills Applied: Embedded Systems, Real-time ML Inference, IoT System Architecture, Autonomous Decision-Making, System Integration.

## EXPERIENCE

- **AI & ML Intern**

Infosys Springboard | Bengaluru, Karnataka (Feb-2026 to April-2026)

Served as a Project Associate in the NLU Bot Trainer program, training and optimizing data pipelines for industrial intent classification. Analyzed natural language dataset patterns to refine intent detection accuracy and model processing logic.

- **Embedded & IoT Intern**

Uni Convergence Technologies | Noida, Uttar Pradesh (May 2025 – Jun 2025)

Prototyped local TinyML inference engines for the CAPHA embedded IoT system, enabling low-latency anomaly detection directly on edge microcontrollers. Developed lightweight sensor processing pipelines operating within strict RAM/Flash memory constraints without external cloud reliance. Collaborated with hardware engineers to validate real-time model execution on low-power IoT devices.

- **Product Intern**

Sri Nikhil Krishna Solutions (SNKS) | Peddapuram, Andhra Pradesh (Nov 2023 – May 2024)

Conducted high-precision electrical characterization and functional quality control procedures for optoelectronic PCB assemblies. Optimized production workflows and testing procedures to improve manufacturing efficiency across multiple assembly phases. Developed structured testing protocols to evaluate hardware quality and minimize initial assembly defects.

## PUBLICATIONS & PREPRINTS

- **Lokesh, G.** (2026). *Early Warning System for At-Risk Students using SHAP values & OULAD dataset*. Preprint available on EdArXiv. DOI: [10.35542/osf.io/ga46q\\_v1](https://doi.org/10.35542/osf.io/ga46q_v1)

Developed and published an open-access predictive framework benchmarking Logistic Regression, Random Forest, and XGBoost across 32,593 subjects and time-series interaction logs. Integrated SHAP (SHapley Additive exPlanations) for transparent root-cause factor attribution, demonstrating advanced capabilities in high-dimensional time-series modeling and model explainability.

## EDUCATION

- **Summer School 2026**

CeNSE, IISc, Bengaluru

**Relevant Coursework:** Semiconductor Fabrication, Computational Lithography, Semiconductor Materials, Masked & Maskless Lithography.

- **Semiconductor Fabrication 101**

Purdue University, UT Austin, and Intel Corporation.

Completed: May 2025

- **Aspire Leadership programme**

Aspire institute.

- **Bachelor of Technology (BTech)** in Electronics & Communication Engineering

Adikavi Nannaya University (AKNU), University College of Engineering, Rajamahendravaram.

Current Status: 3rd Year | Expected Graduation: April 2027

**Relevant Coursework:** Semiconductor Physics, VLSI Technology, Advanced Signal Processing, and Embedded Systems.

- **Nanotechnology: A Maker's Course**

Duke University, NC State University, and UNC Chapel Hill via Coursera.

Completed: Apr 2024

**Relevant Coursework:**

- **Undergraduate Diploma** in Electronics & Communication Engineering

State Board of Technical Education (SBTET) A.P, Government Polytechnic, Anaparthi.

Completed: May 2024

**Relevant Coursework:** Core Electronic Circuits, Microcontroller Architectures, Hardware Testing, and Foundational Engineering Mathematics.

## CERTIFICATIONS

- **PCB Basic Design Course** | Altium Education (Sep 2025)
- **Six Sigma White Belt Certification** | Six Sigma Online / Educate 360 (Dec 2025)
- **Principles of Generative AI** | Infosys Springboard (Dec 2025)
- **Aviatrix Certified Engineer (ACE) – Multicloud Network Associate** | Aviatrix, Inc. (Dec 2025)
- **CompTIA Fundamentals** | Infosys Springboard
- **Forward Program** | McKinsey (Dec 2025)
- **Agile Scrum** | Infosys Springboard